

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – CODING CHALLENGE QUESTIONS

Course Code:	CSA1220	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Coding challenge questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
Student Name:	ASWIN M	Reg. No.:	192412437
Section / Batch:	B.E (ECE) 2024 - 2028	Date:	30/07/2026

Instructions to Students

- Analyze the given scenario and identify the problem requirements before developing the solution.
- Design the algorithm or flowchart and select the appropriate 8085 instructions, registers, and addressing modes required for implementation.
- Develop and execute the 8085 Assembly Language Program (ALP) using a simulator or trainer kit to solve the given problem.
- Verify and validate the output by testing the program with the given input data and ensuring the expected results are obtained.
- Interpret the program execution by explaining the logic used, the role of registers, memory locations, flags, and the final output generated.

QUESTIONS

1. A warehouse stores the quantity of products available in eight storage racks. Before dispatch, the warehouse manager wants the processor to arrange the quantities in ascending order and identify the rack with the lowest stock for replenishment. Write an 8085 ALP to
 - Sort the quantities in ascending order.
 - Store the sorted values.
 - Identify and store the minimum quantity.
2. During data transmission, a block of eight bytes is copied from one memory region to another. To ensure successful transmission, each byte in the destination memory must be compared with its corresponding source byte. Write an 8085 ALP to
 - Copy the data block.
 - Compare each source byte with the copied byte.
 - Count the number of mismatches.
 - Store the mismatch count.
3. A communication controller receives an 8-bit binary code. According to the communication protocol, the code is valid only if it contains an even number of 1s. Develop an 8085 ALP to
 - Count the number of set bits.
 - Determine whether the parity is even or odd.
 - Store a validity flag in memory.

Code of Conduct

I ASWIN M - 192412437 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Marks Evaluation:

Q. No	Problem Understanding (20)	Algorithm (25)	Code Implementation (20)	Competitive Performance (20)	Test case handling (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 20	/ 20	/ 15	/ 100

Course Coordinator

Faculty Incharge

CO1 ASSESSMENT TOOL - 3

Aswin M - 192412437
CSA1220 - Computer Architecture

1. A warehouse stores the quantity of products available in eight storage racks. Before dispatch, the warehouse manager wants the processor to arrange the quantities in ascending order and identify the rack

with the lowest stock for replenishment. Write an 8085 ALP to

- Sort the quantities in ascending order.
- Store the sorted values.
- Identify and store the minimum quantity.

Program Code

```
1  LXI H,8000H
2  MVI C,07H
3
4  LOOP2: MOV D,C
5  LXI H,8000H
6
7  LOOP1: MOV A,M
8  INX H
9  CMP M
10 JC NEXT
11 JZ NEXT
12
13 MOV B,M
14 MOV M,A
15 DCX H
16 MOV M,B
17 INX H
18
19 NEXT: DCR D
20 JNZ LOOP1
21
22 DCR C
23 JNZ LOOP2
24
25 LXI H,8000H
26 MOV A,M
27
28 LXI H,8008H
29 MOV M,A
30
31 HLT
```

After Execution - Sorting

Start	8000	OK
Address (Hex)	Address	Data
1F40	8000	1
1F41	8001	2
1F42	8002	3
1F43	8003	5
1F44	8004	7
1F45	8005	8
1F46	8006	9
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0

Observation

The 8085 Assembly Language Program was written in GNUSim8085 to sort eight product quantities in ascending order. The program also stores the minimum quantity separately after sorting.

Analysis

The processor repeatedly compares two adjacent memory values and exchanges them whenever required. After several passes, all quantities are arranged in ascending order, and the minimum value is obtained from the first memory location and stored separately.

Result

The warehouse quantities were successfully sorted in ascending order, and the minimum quantity was identified and stored separately. This enables the warehouse manager to quickly identify the rack with the lowest stock for replenishment.

2. During data transmission, a block of eight bytes is copied from one memory region to another. To ensure successful transmission, each byte in the destination memory must be compared with its corresponding source byte. Write an 8085 ALP to

- Copy the data block.
- Compare each source byte with the copied byte.
- Count the number of mismatches.
- Store the mismatch count.

Program Code

```
1  LXI H,8000H
2  LXI D,8100H
3  MVI C,08H
4
5  COPY: MOV A,M
6  STAX D
7  INX H
8  INX D
9  DCR C
10 JNZ COPY
11
12 LXI H,8000H
13 LXI D,8100H
14 MVI C,08H
15 MVI B,00H
16
17 COMPARE: MOV A,M
18 LDAX D
19 CMP M
20 JZ NEXT
21
22 INR B
23
24 NEXT: INX H
25 INX D
26 DCR C
27 JNZ COMPARE
28
29 LXI H,8200H
30 MOV M,B
31
32 HLT
```

Before Execution

Start

Address (Hex)	Address	Data
1F40	8000	25
1F41	8001	18
1F42	8002	1
1F43	8003	30
1F44	8004	5
1F45	8005	8
1F46	8006	9
1F47	8007	11
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0

After Execution - Destination Memory

Start

Address (Hex)	Address	Data
1FA4	8100	25
1FA5	8101	18
1FA6	8102	1
1FA7	8103	30
1FA8	8104	5
1FA9	8105	8
1FAA	8106	9
1FAB	8107	11
1FAC	8108	0
1FAD	8109	0
1FAE	8110	0
1FAF	8111	0
1FB0	8112	0
1FB1	8113	0

Analysis

The program first copies the data block from the source to the destination memory. It then compares each source byte with its corresponding destination byte. If a mismatch is detected, the mismatch counter is incremented. Finally, the total mismatch count is stored in memory.

Result

The data block was successfully copied and verified. The mismatch count was stored at memory location **8200H**, confirming the integrity of the transmitted data.

3. A communication controller receives an 8-bit binary code. According to the communication protocol, the

code is valid only if it contains an even number of 1s. Develop an 8085 ALP to

- Count the number of set bits.
- Determine whether the parity is even or odd.
- Store a validity flag in memory.

Program Code

```
1  LDA 8000H
2  MVI B,08H
3  MVI C,00H
4
5  LOOP: RAR
6  JNC NEXT
7  INR C
8
9  NEXT: DCR B
10 JNZ LOOP
11
12 MOV A,C
13 STA 8001H
14
15 ANI 01H
16 JZ EVEN
17
18 MVI A,01H
19 STA 8002H
20 HLT
21
22 EVEN: MVI A,00H
23 STA 8002H
24
25 HLT
```

After Execution

Start	8000	OK
Address (Hex)	Address	Data
1F40	8000	60
1F41	8001	3
1F42	8002	1
1F43	8003	0
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0
1F4E	8014	0

Observation

The total number of set bits was stored in memory location 8001H. A validity flag was stored in 8002H, where 00H indicates even parity (valid) and 01H indicates odd parity (invalid).

Analysis

The processor rotates each bit through the Carry flag and counts the number of set bits. It then checks whether the count is even or odd and stores the corresponding validity flag in memory.

Result

The program successfully counted the set bits, determined the parity of the received 8-bit code, and stored the validity flag in memory according to the communication protocol.